Day 3 – Subqueries, Set Operators & CTEs:

AIRLINES DB

Create Database:

```
Create database AirlinesDB;
Use AirlinesDB;
```

Create Tables:

```
CREATE TABLE air_passenger_profile
(
    profile_id VARCHAR(10) PRIMARY KEY,
    password VARCHAR(10),
    first_name VARCHAR(10),
    last_name VARCHAR(10),
    address VARCHAR(100),
    mobile_number VARCHAR(15),
    email_id VARCHAR(30) UNIQUE
);

CREATE TABLE air_flight
(
    flight_id VARCHAR(10) PRIMARY KEY,
    airline_id VARCHAR(10),
    airline_name VARCHAR(20),
    from_location VARCHAR(20),
    to_location VARCHAR(20),
    departure_time TIME,
    arrival_time TIME,
    duration TIME,
    total_seats INT
);

CREATE TABLE air_flight_details
(
    flight_id VARCHAR(10),
    flight_departure_date DATE,
    price DECIMAL(10,2),
    available_seats INT,
    PRIMARY KEY(flight_id,flight_departure_date),
    CONSTRAINT FK_flight_details
    FOREIGN KEY(flight_id)
    REFERENCES air_flight(flight_id)
);


CREATE TABLE air_ticket_info
(
    ticket_id VARCHAR(10) PRIMARY KEY,
    profile_id VARCHAR(10),
    flight_id VARCHAR(10),
    flight_departure_date DATE,
    status VARCHAR(10),
    CONSTRAINT FK_ticket_profile
    FOREIGN KEY(profile_id)
    REFERENCES air_passenger_profile(profile_id),
    CONSTRAINT FK_ticket_flight
    FOREIGN KEY (flight_id,flight_departure_date)
    REFERENCES air_flight_details(flight_id,flight_departure_date)
```

```sql
);

CREATE TABLE air_credit_card_details
(
    card_number BIGINT PRIMARY KEY,
    profile_id VARCHAR(10),
    card_type VARCHAR(10),
    expiration_month INT,
    expiration_year INT,
    CONSTRAINT FK_prof_cc_details
    FOREIGN KEY(profile_id)
    REFERENCES air_passenger_profile(profile_id)
);
```

Insert Records :

```sql
INSERT INTO air_passenger_profile VALUES
('P201','pwd201','Himapriya','Pothuganti','Hyderabad','9876500001','himapriya@gma
il.com'),
('P202','pwd202','Neha','Reddy','Mumbai','9876500002','neha@gmail.com'),
('P203','pwd203','Amit','Mehta','Ahmedabad','9876500003','amit@gmail.com'),
('P204','pwd204','Vikas','Reddy','Bangalore','9876500004','vikas@gmail.com');

INSERT INTO air_flight VALUES
('F301','AL01','SkyJet','Delhi','Mumbai','06:30','08:45','02:15',180),
('F302','AL02','CloudAir','Hyderabad','Chennai','09:00','10:30','01:30',160),
('F303','AL03','FlyHigh','Bangalore','Kolkata','14:15','17:00','02:45',170),
('F304','AL04','AeroMax','Pune','Delhi','19:00','21:15','02:15',150),
('F401','AL10','ABC','Chennai','Delhi','07:00','09:30','02:30',180),
('F402','AL10','ABC','Mumbai','Hyderabad','13:00','15:00','02:00',160),
('F501','AL20','TestAir','Chennai','Hyderabad','10:00','11:30','01:30',120);

INSERT INTO air_flight_details VALUES
('F301','2025-02-01',5200,140),
('F302','2025-02-01',4500,120),
('F303','2025-02-02',6100,130),
('F304','2025-02-03',5800,110),
('F401','2025-01-10',4800,120),
('F401','2025-01-20',5200,110),
('F401','2025-02-05',5000,115),
('F401','2025-02-25',5400,105),
('F402','2025-01-12',4500,100),
('F402','2025-01-28',4700,95),
('F402','2025-02-10',4900,90),
('F402','2025-02-22',5100,85),
('F501','2025-03-15',4000,100),
('F301','2025-04-10',6300,90);

INSERT INTO air_ticket_info VALUES
('T401','P201','F301','2025-02-01','Booked'),
('T402','P202','F302','2025-02-01','Booked'),
('T403','P203','F303','2025-02-02','Cancelled'),
('T404','P204','F304','2025-02-03','Booked'),
('T501','P201','F501','2025-03-15','Booked'),
('T601','P202','F401','2025-01-10','Booked');

INSERT INTO air_credit_card_details VALUES
(4444333322221111,'P201','VISA',10,2026),
(5555444433332222,'P202','MASTER',6,2025),
(6666555544443333,'P203','VISA',3,2027),
(7777666655554444,'P204','MASTER',9,2026);
```

1. Write a query to display the average monthly cket cost for each flight in ABC Airlines. The query should display the Flight_Id, From_loca on, To_Loca on, Month Name as “Month_Name” and average price as “Average_Price”. Display the records sorted in ascending order based on flight id and then by Month Name.

```
select avg(afd.price),af.flight_id,af.from_location,af.to_location,datename
(MONTH,afd.flight_departure_date)as Month_name
from air_flight af
join air_flight_details afd
on afd.flight_id=af.flight_id
WHERE af.airline_name = 'ABC'
group by af.flight_id,af.from_location,af.to_location,datename
(MONTH,afd.flight_departure_date) ,MONTH(afd.flight_departure_date)
order by af.flight_id,MONTH(afd.flight_departure_date);
```

	avg_price	flight_id	from_location	to_location	Month_name
1	5000.000000	F401	Chennai	Delhi	January
2	5200.000000	F401	Chennai	Delhi	February
3	4600.000000	F402	Mumbai	Hyderabad	January
4	5000.000000	F402	Mumbai	Hyderabad	February

2. Write a query to display the customer(s) who has/have booked least number of ckets in ABC Airlines. The Query should display profile_id, customer’s first_name, Address and Number of ckets booked as “No_of_Tickets”. Display the records sorted in ascending order based on customer's first name.

```
select ap.profile_id, ap.first_name, ap.address,
       count(ati.ticket_id) as No_of_Tickets
from air_ticket_info as ati
join air_passenger_profile as ap
on ap.profile_id = ati.profile_id
join air_flight_details as afd
on afd.flight_id = ati.flight_id
join air_flight as af
on af.flight_id = afd.flight_id
group by ap.profile_id, ap.first_name, ap.address
having count(ati.ticket_id) = (
    select min(ticket_count)
    from (
        select count(ati.ticket_id) as ticket_count
        from air_ticket_info as ati
        join air_passenger_profile as ap
        on ap.profile_id = ati.profile_id
        join air_flight_details as afd
        on afd.flight_id = ati.flight_id
        join air_flight as af
        on af.flight_id = afd.flight_id
        group by ap.profile_id
    ) as min_ticket_count
)order by ap.first_name;
```

	profile_id	first_name	address	No_of_Tickets
1	P203	Amit	Ahmedabad	1
2	P204	Vikas	Bangalore	1

3. Write a query to display the number of flight services between locations in a month. The Query should display From_Location, To_Location, Month as "Month_Name" and number of flight services as "No_of_Services". The records should be displayed in ascending order based on From_Location, To_Location and Month Name.

```
select af.from_location, af.to_location,
       datename(month, fd.flight_departure_date) as Month_Name,
       count(af.flight_id) as No_of_Services
from air_flight_details as fd
join air_flight as af
on fd.flight_id = af.flight_id
group by af.from_location, af.to_location,
         datename(month, fd.flight_departure_date)
order by af.from_location, af.to_location, Month_Name;
```

	from_location	to_location	Month_Name	No_of_Services
1	Bangalore	Kolkata	February	1
2	Chennai	Delhi	February	2
3	Chennai	Delhi	January	2
4	Chennai	Hyderabad	March	1
5	Delhi	Mumbai	April	1
6	Delhi	Mumbai	February	1
7	Hyderabad	Chennai	February	1
8	Mumbai	Hyderabad	February	2
9	Mumbai	Hyderabad	January	2
10	Pune	Delhi	February	1

4.Write a query to display the customer(s) who has/have booked maximum number of tickets in ABC Airlines. The Query should display profile_id, customer's first_name, Address and Number of Tickets booked as "No_of_Tickets". Display the records sorted in ascending order based on customer's first name

```
select ap.profile_id, ap.first_name, ap.address,
       count(ati.ticket_id) as No_of_Tickets
from air_ticket_info as ati
join air_passenger_profile as ap
on ap.profile_id = ati.profile_id
join air_flight_details as afd
on afd.flight_id = ati.flight_id
join air_flight as af
on af.flight_id = afd.flight_id
group by ap.profile_id, ap.first_name, ap.address
having count(ati.ticket_id) = (
    select max(ticket_count)
    from (
        select count(ati.ticket_id) as ticket_count
        from air_ticket_info as ati
        join air_passenger_profile as ap
        on ap.profile_id = ati.profile_id
        join air_flight_details as afd
        on afd.flight_id = ati.flight_id
        join air_flight as af
        on af.flight_id = afd.flight_id
        group by ap.profile_id
    ) as max_ticket_count
```

```
)
order by ap.first_name;
```

	profile_id	first_name	address	No_of_Tickets
1	P203	Amit	Ahmedabad	1
2	P204	Vikas	Bangalore	1

5.Write a query to display the number of ckets booked from Chennai to Hyderabad. The Query should display passenger profile_id,first_name,last_name, Flight_Id , Departure_Date and number of ckets booked as “No_of_Tickets”. Display the records sorted in ascending order based on profile id and then by flight id and then by departure date.

```
select
    pf.profile_id,
    pf.first_name,
    pf.last_name,
    ti.flight_id,
    ti.flight_departure_date,
    Count(ti.ticket_id) as no_of_tickets
from air_passenger_profile pf
join air_ticket_info  ti on ti.profile_id=pf.profile_id
join air_flight f on f.flight_id=ti.flight_id
where f.from_location='Chennai' and f.to_location ='Hyderabad'
group by
pf.profile_id,pf.first_name,pf.last_name,ti.flight_id,ti.flight_departure_date
order by pf.profile_id,ti.flight_id,ti.flight_departure_date;
```

	profile_id	first_name	last_name	flight_id	flight_departure_date	no_of_tickets
1	P201	Himapriya	Pothuganti	F501	2025-03-15	1

6. Write a query to display flight id,from location, to location and ticket price of flights whose departure is in the month of april.

```
select
    fd.flight_id,
    f.from_location,
    f.to_location,
    fd.price
from air_flight_details fd
join air_flight f
on f.flight_id=fd.flight_id
where datename (MONTH,fd.flight_departure_date) ='April';
```

	flight_id	from_location	to_location	price
1	F301	Delhi	Mumbai	6300.00

7.Write a query to display the average cost of the tickets in each flight on all scheduled dates. The query should display flight_id, from_location, to_location, and average price as “Price”. Display the records sorted in ascending order based on flight_id, then by from_location, and then by to_location.

```
select
    f.flight_id,
```

```
    f. from_location,
     f.to_location,
     AVG(fd.price) as avg_price

From air_flight f
Join air_flight_details fd
    ON f.flight_id = fd.flight_id

group by f.flight_id,f.from_location,f.to_location
order by f.flight_id,f.from_location,f.to_location;
```

	flight_id	from_location	to_location	avg_price
1	F301	Delhi	Mumbai	5750.000000
2	F302	Hyderabad	Chennai	4500.000000
3	F303	Bangalore	Kolkata	6100.000000
4	F304	Pune	Delhi	5800.000000
5	F401	Chennai	Delhi	5100.000000
6	F402	Mumbai	Hyderabad	4800.000000
7	F501	Chennai	Hyderabad	4000.000000

8.Write a query to display the customers who have booked tickets from Chennai to Hyderabad. The query should display profile_id, customer_name (combine first_name and last_name with a comma in between), and address of the customer.
Give an alias to the name as customer_name.
Hint: The query should fetch unique customers irrespective of multiple tickets booked.
Display the records sorted in ascending order based on profile_id.

```
select
    pf.profile_id,
    concat(pf.first_name,pf.last_name) as customer_name,
    pf.address
from air_passenger_profile pf
join air_ticket_info  ti on ti.profile_id=pf.profile_id
join air_flight f on f.flight_id=ti.flight_id
where f.from_location='Chennai' and f.to_location ='Hyderabad'

order by pf.profile_id;
```

	profile_id	customer_name	address
1	P201	HimapriyaPothuganti	Hyderabad

9.Write a query to display profile id of the passenger(s) who has/have booked the maximum number of tickets.
In case of multiple records, display the records sorted in ascending order based on profile_id.

```
select profile_id
from  air_ticket_info
group by  profile_id
having count(ticket_id) = (
        select max(cnt)
        from (
            select count(ticket_id) AS cnt
            from air_ticket_info
            group by profile_id
        ) t
)
```

```sql
ORDER BY profile_id;
```

	profile_id
1	P201
2	P202

10.Write a query to display the total number of tickets as "No_of_Tickets" booked in each flight in ABC Airlines.
The query should display flight_id, from_location, to_location, and the number of tickets.
Display only those flights in which at least one ticket is booked.
Display the records sorted in ascending order based on flight_id.

```sql
select
    f.flight_id,
    f.from_location,
    f.to_location,
    count(t.ticket_id) as no_of_tickets
from air_flight f
join air_ticket_info t
    on f.flight_id = t.flight_id
where f.airline_name = 'ABC'
group by
    f.flight_id,
    f.from_location,
    f.to_location
having count(t.ticket_id) >= 1
order by f.flight_id asc;
```

	flight_id	from_location	to_location	no_of_tickets
1	F401	Chennai	Delhi	1